



Libbie Henrietta Hyman

*L.H. Hyman is a very big name in 'Invertebrate Zoology'. Her six volumes on the systematics of the invertebrates are monumental editions - a work which goes in **gilt letters** in the annals of the History of Zoology.*

Unit

3

Animal Diversity - I

Non - Chordate Phyla

LIFE - LOOKS SO SIMPLE - YET IT IS SO COMPLEX

ANIMAL DIVERSITY, THE LANGUAGE OF COMPLEXITY

Animalia (**Metazoa**) includes **multi-cellular**, **heterotrophic** (obtain nourishment by mostly ingesting and digesting the ingested food), **eukaryotes** with specific body plans (**bauplans**). Animal cells do not have cellulose cell walls, but they are held together by certain structural proteins. **Collagen** is the most abundant protein and unique to the Kingdom Animalia.

Animals reproduce mostly sexually, by the fusion of 'gametes' resulting in the production of '**zygotes**'. Zygotes undergo different types of cleavages and pass through simple stages such as '**blastula**', '**gastrula**' etc. and further differentiate into an adult, either '**directly**' (without a larval stage), or '**indirectly**' (with a larval stage).

The most simple multi-cellular organisms are the 'sponges' which show '**cellular level of organisation**'. The 'choano-flagellates' (mastigophores) of the kingdom '**Protista**'/ '**Protoctista**', are the closest **living relatives** of the metazoans. While sponges have 'unorganised' cells, the 'diploblastic organisms' (cnidarians and ctenophores), are the 'first true metazoans' with two germinal layers, namely 'ectoderm' and 'endoderm'. They show '**tissue level of organisation**'. The diploblastic organisms show 'co-ordination' between body parts, due to the presence of '**sensory cells**' and '**nerve cells**'.

The next higher level of organisation is reached with the development of the 'triploblastic level of organisation' (three germinal layers namely – **ectoderm**, **endoderm** and the third germinal layer, the **mesoderm**) – a promotion from '**tissue**' to

‘organ’ and ‘organ-system’ level of organisation. The first ‘triploblasts’ have a ‘**solid bauplan**’ (the space between the ectoderm and endoderm/ the perivisceral space, is filled with a tissue called ‘parenchyma/mesenchyma’ derived from the ‘**mesoderm**’ - the ‘**Acoelomates**’).

The first ‘**tube - within - a tube**’ body organisation (outer tube-the **body wall** and the inner tube - the **alimentary canal**) arose with the evolution of the nematodes. The space between the body wall and the alimentary canal, in the body of a nematode is the remnant of the ‘primary body cavity’ (**blastocoel**) and is called ‘**pseudocoel**’ (false coelom). Coelom (perivisceral space lined by mesodermal coelomic epithelia/ peritoneal layers) arose by the splitting of the mesoderm. It is called ‘**schizocoelom**’ (schizo- splitting). It is typically seen in the ‘annelids’, ‘arthropods’ and ‘molluscs’. The echinoderms, hemichordates and chordates also have a true coelome (**eucoelom**). It arises by the confluence of the ‘out-pouchings’ of the archenteron (primitive gut). As such the coelom in them is described as ‘**enterocoelom**’.

In the development of animals, the blastula stage develops into gastrula stage. Gastrula is a ‘double layered stage’ with a new cavity called ‘**archenteron**’, which opens to the exterior by a ‘**blastopore**’. It is a very crucial in the developmental processes. Gastrulation finally leads to the differentiation of the three embryonic cell layers (**germinal layers**). The organisms in which the blastopore develops into the **mouth** are called ‘**protostomes**’ (**platyhelminths** to the **molluscs**) and the organisms in which the blastopore usually develops into the ‘**anus**’ and the ‘mouth’ is formed secondarily are called ‘**deuterostomes**’ e.g., **echinoderms**, **hemichordates** and **chordates**. In the protostomes cleavage is ‘**spiral**’ and ‘**determinate**’. The embryos of protostomes are described as ‘**mosaic embryos**’. The cleavage in the deuterostomes is ‘**radial**’ and ‘**indeterminate**’ and their embryos are called ‘**regulative embryos**’. The fate of each blastomere is ‘**predetermined**’ in the case of determinate development and in the case of regulative embryos, their fate is decided late in the development.

Evolution of tissues, development of **bilateral symmetry** and origin of the **body cavity** are successively considered the **first**, **second** and the **third** ‘**key transitions**’ in the animal organisation. This **Unit** traces the entire history of evolution of organisation in the animal kingdom from simple to more complex levels. You will be learning more **salient features** of different major groups (**phyla**) of the Kingdom **Animalia** as you are taken from the study of the **sponges** to that of the **chordates**.

Animals are broadly classified into **invertebrates** and **vertebrates**. The animals which lack a **notochord** are called **invertebrates**. **Libbie Henrietta Hyman** (1888-1969) was the ‘pioneer’ in the field of systematics of the invertebrates. Her legendary volumes (6 volumes, 1940-68) on the **Invertebrates** are popular and considered authoritative, even in recent times. The invertebrates are characterized by the absence of vertebral column (back bone) and presence of solid, ventral ganglionated nerve cord. Alimentary canal is dorsal to the nerve cord. They exhibit many types of reproduction.

Invertebrate Phyla

- 3.1 Phylum: Porifera
- 3.2 Phylum: Cnidaria
(Coelenterata)
- 3.3 Phylum: Ctenophora
- 3.4 Phylum: Platyhelminthes
- 3.5 Phylum: Nematoda
- 3.6 Phylum: Annelida
(Earthworm – Type study)
- 3.7 Phylum: Arthropoda
- 3.8 Phylum: Mollusca
- 3.9 Phylum: Echinodermata
- 3.10 Phylum: Hemichordata



3.1 PHYLUM-PORIFERA

(Subkingdom-Parazoa)

- An Evolutionary Blind Offshoot

3.1.1 General Characters

The term *Porifera* (**pore bearing**) was coined and their true ‘animal nature’ was established by **Robert Grant**. Members of this phylum are commonly called sponges. They are generally marine (with the exception of families Potamolepidae and Spongillidae, which include freshwater **sponges** e.g. *Spongilla*). They are either **radially symmetrical** or **asymmetrical** animals (Figure 3.1). These are primitive, **multicellular** and sessile animals and have **cellular level of organisation**. The body wall is composed of two layers (outer epidermis/ pinacoderm and inner choanoderm) separated by a gelatinous matrix called **mesohyl**. Sponges have a **canal system** that constantly conducts water. Water enters through minute pores called **ostia** (inhalent apertures) in the body wall into a central cavity, the **spongocoel** (paragastric cavity), from where it goes out through the **osculum** (exhalent aperture). This transport of water is helpful in gathering food (nutrition-**filter feeders**), respiratory exchange of gases (respiration) and removal of wastes (excretion). **Choanocytes** or **collar cells** line the spongocoel and the canals. Nutrition is holozoic and digestion is ‘**intracellular**’. Reserve food is stored in cells called ‘**thesocytes**’. Nerve cells and sensory cells are absent and so they do not show much co-ordination between the functioning of various parts of the body. The body is supported by a skeleton made up of **calcareous or siliceous spicules or spongin fibres** or both. Sexes are not separate (hermaphroditic or monoecious), i.e., egg cells and sperms are produced by the same individual. Sponges reproduce asexually by fragmentation, budding etc., and sexually by formation of gametes.

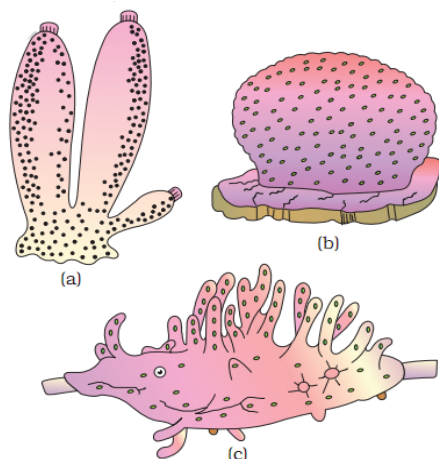


Fig 3.1 Examples for Porifera :
(a) *Sycon* (b) *Euspongia* (c) *Spongilla*

Fertilisation is internal. Cleavage is holoblastic and development is indirect involving different types of larval stages such as **parenchymula**, **trichimella** and **amphiblastula** which are morphologically different from the adult. Power of regeneration is well developed. They are considered an evolutionary blind offshoot.

Examples: *Sycon*, *Euplectella*, *Spongilla* and *Euspongia*.

3.1.2 Classification of the phylum Porifera

Phylum Porifera is classified into three classes,
1. Calcarea 2. Hexactinellida and **3. Demospongiae**

Class-1: Calcarea

They live in shallow marine waters. They are solitary or colonial. Body structure is simple. Spicules are calcareous

(made up of Ca CO_3).

Examples: *Sycon* (*Scypha*), *Leucosolenia*, *Grantia*.

Class-2: Hexactinellida

They live in deeper parts of the sea. They are solitary. Spicules are six rayed (hexactinal) and siliceous (made of silicon dioxide – *glass*).

Examples: *Euplectella* (Venus' flower basket), *Hyalonema* (Glass - rope sponge).

Class-3: Demospongiae

There are both marine and freshwater sponges in this class. They are colonial. Skeleton consists of siliceous spicules (other than six rayed) or spongin fibres or both or skeletal structures may be absent.

Examples: *Spongilla* (freshwater sponge), *Euspongia* (bath sponge), *Chalina* (dead man's fingers).

3.2 PHYLUM-CNIDARIA (COELENTERATA)

- *Tissues and the radial body plan*

3.2.1 General Characters



Cnidarians (previously called coelenterates) are aquatic, mostly marine, solitary or colonial, sessile or free-swimming, **radially symmetrical** animals (sea anemones are biradially symmetrical) (Figure 3.2). The recent name Cnidaria is derived from the '**stinging cells**' called **cnidoblasts** or **cnidocytes** present mostly on the tentacles and the body. Cnidoblasts are used for anchorage, defence and capture of prey (Figure 3.3). Cnidarians are the first metazoans to exhibit '**tissue level of organisation**' and are '**diploblastic**'. They have a central

gastro-vascular cavity or **coelenteron** (serves both digestive and circulatory functions), hence the earlier name coelenterata, with a single opening, the mouth serving the purpose of ingestion and egestion, there being no anus. Digestion is both **extracellular** (in the coelenteron) and **intracellular** (in the nutritive muscular cells of the endoderm). Nervous system is of primitive type formed of diffuse '*nerve net*' (formed by non-polarised nerve cells). Sensory structures such as **statocysts** occur in the medusoid forms. Some of the cnidarians, e.g., coral forming cnidarians have an exoskeleton composed of calcium carbonate. Cnidarians show two basic body forms called **polyp** and **medusa** (Figure 3.2). The former is a sessile and cylindrical form (e.g. *Hydra*, *Adamsia* etc.), whereas, the latter is '*umbrella-shaped*' and free-swimming form (e.g. *Aurelia* – commonly called jelly fish). Those cnidarians which exist in both forms exhibit alternation of generations (**metagenesis**), i.e., polypoid forms produce medusae asexually and medusae form the polypoid forms sexually (e.g. *Obelia*). Asexual reproduction is by budding and sexual reproduction is by syngamy. Development is indirect and includes a free swimming ciliated larva, the **planula**.

Examples: *Physalia*, *Adamsia*, *Pennatula*, *Rhizostoma* and *Meandrina*.

3.2.2 Classification of the phylum Cnidaria (Coelenterata)

Phylum Cnidaria is classified into three classes, namely **1. Hydrozoa**, **2. Scyphozoa** and **3. Anthozoa**.

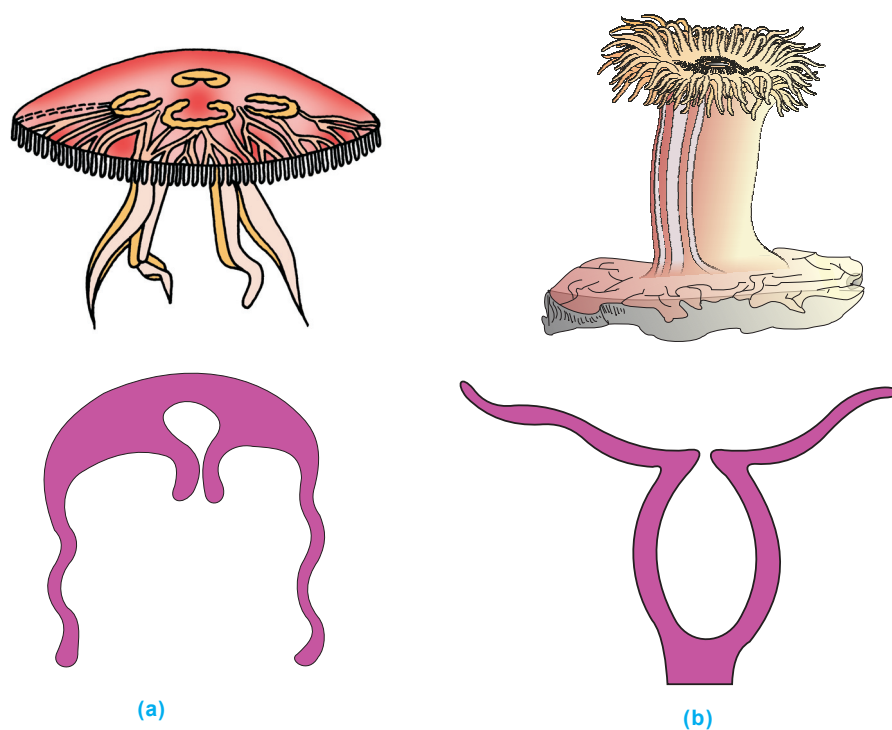


Fig. 3.2 Examples of coelenterata indicating outline of their body form:
(a) *Aurelia* (medusa) (b) *Adamsia* (polyp)

Class-1: Hydrozoa

They may be solitary (e.g. *Hydra*) or colonial. Life history includes sedentary, asexual, polypoid form and free swimming, sexual, medusoid form alternating with each other (e.g. *Obelia*). Mesoglea is non-cellular. Coelenteron is undivided (simple). Cnidocytes (cnidoblasts) occur only in the ectoderm. Germ cells are derived from the ectoderm. Some exhibit *polymorphism* with different types of **zooids** e.g. *Physalia*.

Examples: *Hydra* (freshwater polyp), *Obelia* (sea fur), *Physalia* (Portuguese man-of-war). Note: Zooid is a single individual in a colony.

Do you know why *Physalia* is commonly called ‘Portuguese man-of-war’ or ‘man o’ war’? The word ‘Man of War’ means a ‘battle ship with cannons and other powerful weapons’ Portugal once had. *Physalia* has many tentacles provided with venomous ‘stinging cells’, that are harmful to other organisms, hence the name ‘Portuguese man-of-war’.

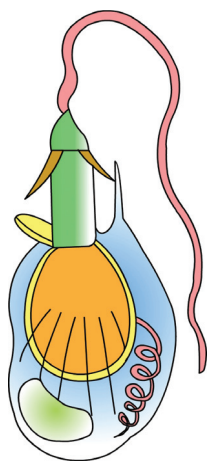


Fig. 3.3 Diagrammatic view of cnidoblast

Class-2: Scyphozoa

They are commonly referred to as jelly fishes as they possess a large quantity of jelly like mesoglea. They are solitary and medusoid in form. Mouth is surrounded by four oral arms. Coelenteron is divided into four chambers and some canals. Mesoglea contains amoebocytes. Cnidocytes occur in ectoderm and endoderm. Germ cells are derived from the endoderm.

Examples: *Aurelia* (jelly fish or moon jelly), *Rhizostoma*.

Class-3: Anthozoa or Actinozoa

They are commonly referred to as *sea anemones*. They are sedentary and *only* polypoid in form. Coelenteron is divided into several compartments by vertical septa called mesenteries. Mesoglea contains connective tissue. Cnidocytes occur both in the ectoderm and endoderm. Germ cells are derived from the endoderm.

Examples: *Adamsia* (sea anemone), *Corallium rubrum* (precious red stone coral), *Gorgonia* (sea fan), *Pennatula* (sea pen).

3.3 PHYLUM-CTENOPHORA

- *Close relatives of cnidarians*

3.3.1 General Characters

Ctenophores, commonly known as ‘**sea walnuts**’ or ‘**comb jellies**’ or ‘**sea gooseberries**’, are exclusively marine, **radially symmetrical**, diploblastic organisms with tissue level of organisation. The body bears *eight* external rows of ciliated **comb plates**, which help in locomotion, hence the name ‘ctenophora’. Cnidocytes are absent. However, they possess ‘**glue cells**’ called



lasso cells or **colloblasts** which help in food capture. Digestion is both extracellular and intracellular. **Bioluminescence** (the property of emitting light by living organism) is well-marked in the ctenophores. Sexes are not separate (monoecious). Reproduction takes place only by sexual method. Fertilisation is external. Development is indirect and includes a larval stage called **cydippid larva**.

Examples: *Pleurobrachia*, *Hormiphora*.

3.3.2 Classification of the phylum Ctenophora

Phylum Ctenophora is classified into two classes,

1. Tentaculata and 2. Nuda.

Class-1: Tentaculata

The adults possess two aboral tentacles.

Examples: *Pleurobrachia*, *Ctenoplana*.

Class-2: Nuda

Tentacles are absent even in the larval stages.

Examples: *Beroë*.

Note: **Sea anemones and sea walnuts are said to be 'Biradially Symmetrical'.**

3.4 PHYLUM-PLATYHELMINTHES

- **Mesoderm and the bilateral plan**

3.4.1 General Characters

They have **dorso-ventrally flattened body**, hence are called **flatworms**. These are mostly endoparasites found in animals, including human beings. Flatworms are the first **bilaterally symmetrical**, **triploblastic** and **acoelomate** animals with **organ-system level of organisation** (Fig. 3.6-3.8). Body is not segmented, but some exhibit pseudometamerism. They show moderate cephalization and unidirectional movement (locomotion). **Hooks** and **suckers** are present in the parasitic forms (Fig. 3.5). Digestive system, if present, has mouth only; anus is absent. Nervous system is moderately developed with brain (cephalic ganglia) and nerve cords forming 'ladder like' system. Specialised excretory cells called **flame cells (protonephridia)** help in osmoregulation and excretion. Sexes are not separate (monoecious). Fertilisation is internal and development is indirect with many stages (**miracidium**, **sporocyst**, **redia**, **cercaria** etc.). **Polyembryony** is common in some (Liver

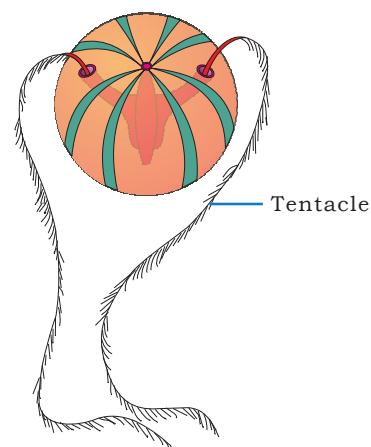


Fig. 3.4 Example of ctenophora (*Pleurobrachia*)

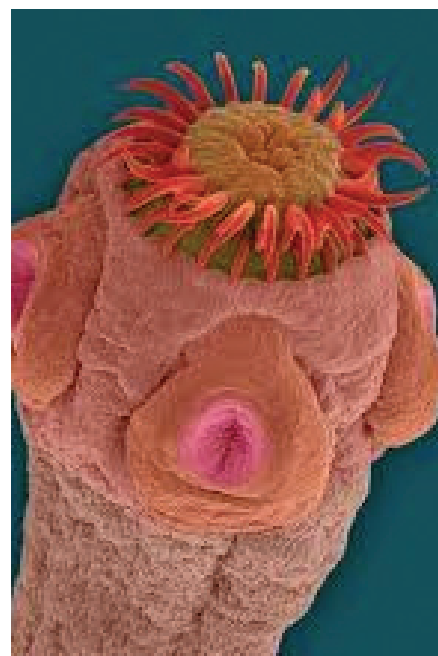


Fig. 3.5 Scolex of tape worm

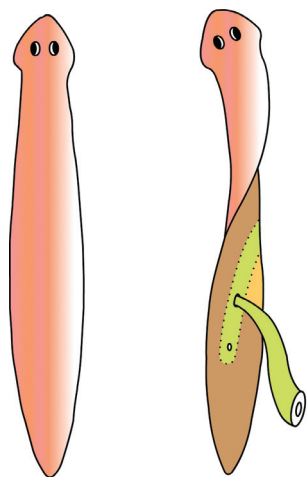


Fig. 3.6 Dugesia

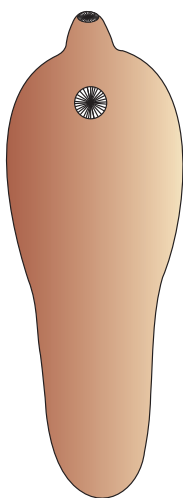


Fig. 3.7 Fasciola

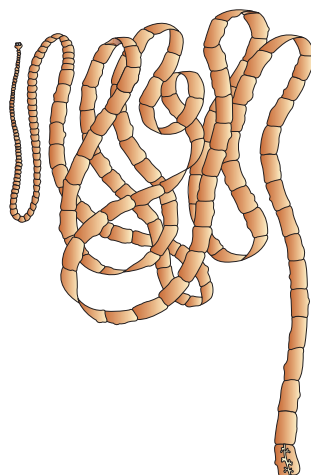


Fig. 3.8 Taenia

fluke). Some members like *Planaria* possess high degree of **regeneration** capacity.

Examples: *Taenia*, *Fasciola*, *Schistosoma*, *Dugesia*.

3.4.2 Classification of the phylum Platyhelminthes

Phylum Platyhelminthes is classified into three classes, namely **1. Turbellaria** **2. Trematoda** and **3. Cestoda**.

Class-1: Turbellaria

They are mostly free living, unsegmented flat worms. Body wall bears cilia and mucus forming structures called **rhabdites** in the epidermal cells. Reproduction is by asexual and sexual methods. Development is indirect and includes free swimming, ciliated **Muller's larva**.

Examples: *Dugesia*, *Convoluta*.

Class-2: Trematoda

They are commonly called *flukes*. They are parasitic on other animals. Body is covered by a thick cuticle (tegument); bears two suckers, an oral and a ventral (acetabulum). Mouth is anterior and the intestine is bifurcated. They are bisexual (monoecious). Life history is complex with many hosts and different types of stages (*miracidium*, *sporocyst*, *redia*, *cercaria* etc.).

Examples: *Fasciola* (liver fluke), *Schistosoma* or *Bilharzia* (blood fluke).

Class-3: Cestoda

They are commonly called *tapeworms*. All are endoparasites. Body is covered by protective **syncytial tegument**. They show **pseudometamerism** (body is divided into segment like units called proglottids which are not 'true segments'). Scolex bears hooks, suckers and **bothridia** (leaf like outgrowths) for anchoring. Mouth and gastrovascular cavity are absent. They are bisexual (monoecious). Life history is complex with many hosts and development is indirect with cysticercus larva (bladder worm).

Examples: *Taenia solium* (pork tapeworm), *Echinococcus granulosus* (dog tapeworm).

3.5 PHYLUM-NEMATODA

(*Aschelminthes*, as used in NCERT book)

- Body cavity, a 'one way gut' and 'tube with-in-a tube'

3.5.1 General Characters

The body of a nematode is circular in cross-section, hence the name **s** (Figure 3.9). They may be free living or parasitic on plants and animals, aquatic or terrestrial. Roundworms have organ-system level of organisation. They are bilaterally symmetrical, triploblastic and **pseudocoelomate** animals. Body is unsegmented and covered by a transparent, tough and protective **collagenous cuticle** (a *unique character*). In some, the epidermis is **syncytial**. Alimentary canal is complete with a mouth, well developed muscular pharynx and anus. Excretory system consists of **renette gland** (excretory gland) and ducts. Nervous system consists of a circum enteric nerve ring with ganglia and nerves. Sense organs like **amphids** and **phasmids** occur. Sexes are separate (dioecious). They exhibit sexual dimorphism, often females are longer and larger than males. Males have curved posterior end with a cloacal aperture and one or two copulatory (penial) spicules. Fertilisation is internal, majority are oviparous (e.g. *Ascaris*), a few are ovo-viviparous (e.g. *Wuchereria*) and development may be direct (the young ones resemble the adult- called **juveniles**) or indirect. Growth into adult forms involves typically four **moultings** in the larval or juvenile stages.

Examples: *Ascaris*, *Wuchereria*, *Ancylostoma*.

In nematodes, the number of body cells (except the cells in gonads) remains constant, once the animal reaches the adulthood. It means the number of nuclei too is constant. This phenomenon is called 'eutely'. If there is growth in the size of the body, it is due to the growth in the size of the cells (hypertrophy).

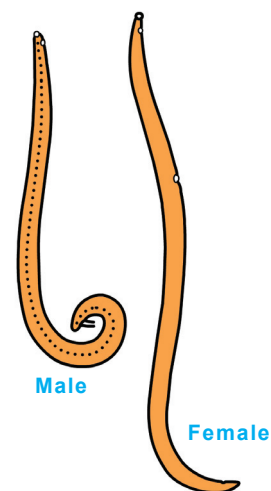


Fig. 3.9 *Ascaris lumbricoides*

3.5.2 Classification of the phylum Nematoda

Phylum Nematoda is classified into two classes, namely **1. Aphasmidia** and **2. Phasmidia**.

Class-1: Aphasmidia

In the members of this class **amphids** (cuticular depressions around oral region performing chemoreceptor function) are highly modified, but phasmids (posterior glandulo-sensory structures) are absent.

Examples: *Trichinella* (trichinaworm), *Trichuris* (whipworm).

Class-2: Phasmidia

In the members of this class amphids are simple and **phasmids** are present. Excretory system is well developed.

Example: *Ascaris* (roundworm), *Ancylostoma* (hookworm), *Enterobius* (pinworm), *Wuchereria* (filarial worm).

3.6 PHYLUM-ANNELIDA

-Segmented body plan

3.6.1 General Characters

The term annelida (L. *annulus* : little ring and Gr. *edios* : form) was coined by **Lamarck**. They may be aquatic (marine and freshwater) or terrestrial; free-living, and sometimes parasitic. They exhibit organ-system level of body organisation and bilateral symmetry. They are triploblastic, **metamerically segmented (metamerism)** and **coelomate (schizocoelic coelom)** animals (Fig. 3.10-3.12). Their body and the coelom are divided by transverse septa into segments or **metameres**. They possess longitudinal and circular muscles in the body wall, which help in locomotion. Coelom with coelomic fluid provides a **hydrostatic skeleton** giving stiffness to the body parts thus aiding in locomotion. Aquatic annelids such as *Nereis* possess lateral appendages, the **parapodia**, which help in swimming; chitinous *setae* and *suckers* also help in locomotion in some. Cephalization is more pronounced with distinct head and sense organs in some polychaetes. Respiration is by simple diffusion through the body wall and in *Nereis* highly vascularised parapodia help in respiration. A *closed* circulatory system is present. Respiratory pigments such as **haemoglobin**, **chlorocruorin** are found dissolved in the plasma (a feature unique to the invertebrates). Nephridia (metanephridia- nephridia with nephrostomes), which are 'ectodermal' in origin, help in osmoregulation and excretion. Nervous system consists of paired (cerebral and sub-pharyngeal) ganglia connected by lateral nerves around the pharynx forming a **nerve ring** and to a ganglionated 'double ventral nerve cord' (annelidan type of nervous system). Sense organs like eyes, palps and tentacles occur in some. They may be dioecious (sexes are separate-e.g. *Nereis*) or monoecious (hermaphrodites – e.g. earthworms and leeches). Reproduction is usually sexual. Cleavage is holoblastic (complete) and spiral. Development is direct in the monoecious (bisexual) annelids and indirect in the dioecious (unisexual) annelids. Characteristic larval form, the **trochophore** occurs in the life history.

Examples: *Nereis*, *Pheretima*, *Megascolex* and *Hirudinaria*.

3.6.2 Classification of the phylum Annelida

Phylum Annelida is classified into three classes, namely

1. Polychaeta 2. Oligochaeta 3. Hirudinea

Class-1: Polychaeta

Polychaeta (*poly* : many; *chaetae* : setae) includes marine annelids commonly called bristle worms. Some are free-moving and others burrowing or tubicolous (tube dwelling). Head is distinct with sense organs such as eyes, tentacles and palps. **Parapodia** which bear many setae (hence the name polychaeta) help in locomotion and respiration. Clitellum is absent. They are dioecious (unisexual); gonoducts are absent; gametes are shed into the coelom and passed out through the nephridiopores. Fertilization is external; development includes a trochophore larva.

Examples: *Nereis* (sandworm or ragworm or clam worm), *Aphrodite* (sea mouse), *Arenicola* (lugworm).

Class-2: Oligochaeta

Oligochaeta (*oligos* : few; *chaetae* : setae) includes earthworms which live in moist soil and some live in freshwater (e.g. *Tubifex*). They lack distinct head. Chitinous setae help in locomotion. Clitellum forms a cocoon during the breeding season. They are monoecious (bisexual); development is direct without larval stage.

Examples: *Pheretima*, *Megascolex* and *Tubifex*.

Class-3: Hirudinea

Hirudinea (*hirudo* : leech) includes *leeches*; all are ectoparasites; majority live in fresh water; some are marine and others live on moist land (terrestrial). They have dorso-ventrally flattened body with a **definite number** of segments ; the segments are externally subdivided into **annuli**; internal segmentation is absent ; suckers help in locomotion. Clitellum is conspicuous during the breeding season only. Coelom is filled with a characteristic tissue called **botryoidal tissue** (*botryoidal*: resembling a bunch of grapes; opinions differ on the function of botryoidal tissue - they range from excretion to storage of iron, calcium, revascularization in areas of injury etc.). They are monoecious (hermaphroditic); male reproductive system possesses a copulatory organ, the *cirrus*. Fertilization is internal and development is direct .

Examples: *Hirudinaria* (freshwater leech), *Pontobdella* (marine leech), *Haemadipsa* (land leech).

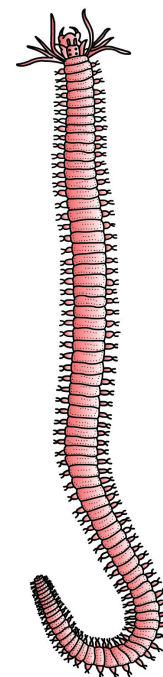


Fig. 3.10 Nereis



Fig. 3.11 Leech

3.6.3 EARTHWORM



Fig. 3.12 Earthworm

Earthworm is a reddish brown terrestrial invertebrate that inhabits the upper layers of the moist soil rich in decaying organic matter. It is nocturnal and during the day time, it lives in burrows made by boring and swallowing the soil. In gardens, they can be traced by their faecal deposits known as **worm castings**. They are considered, the **Friends of farmers**. The common Indian earthworms are *Pheretima*, *Lumbricus* and *Megascolex*. ***Drawida grandis*** of south India is the longest earthworm in India. *Chaetogaster annandalei* is the smallest earthworm, while *Megascolides australis* is the longest earthworm.

I. Morphology

Pheretima posthuma has a long cylindrical body which is divided into 100 to 120 segments (metameres). The dorsal surface of the body is marked by a dark mid-dorsal line (representing the dorsal blood vessel) along the longitudinal axis of the body. The ventral surface of the body shows genital apertures (pores). The anterior end bears a

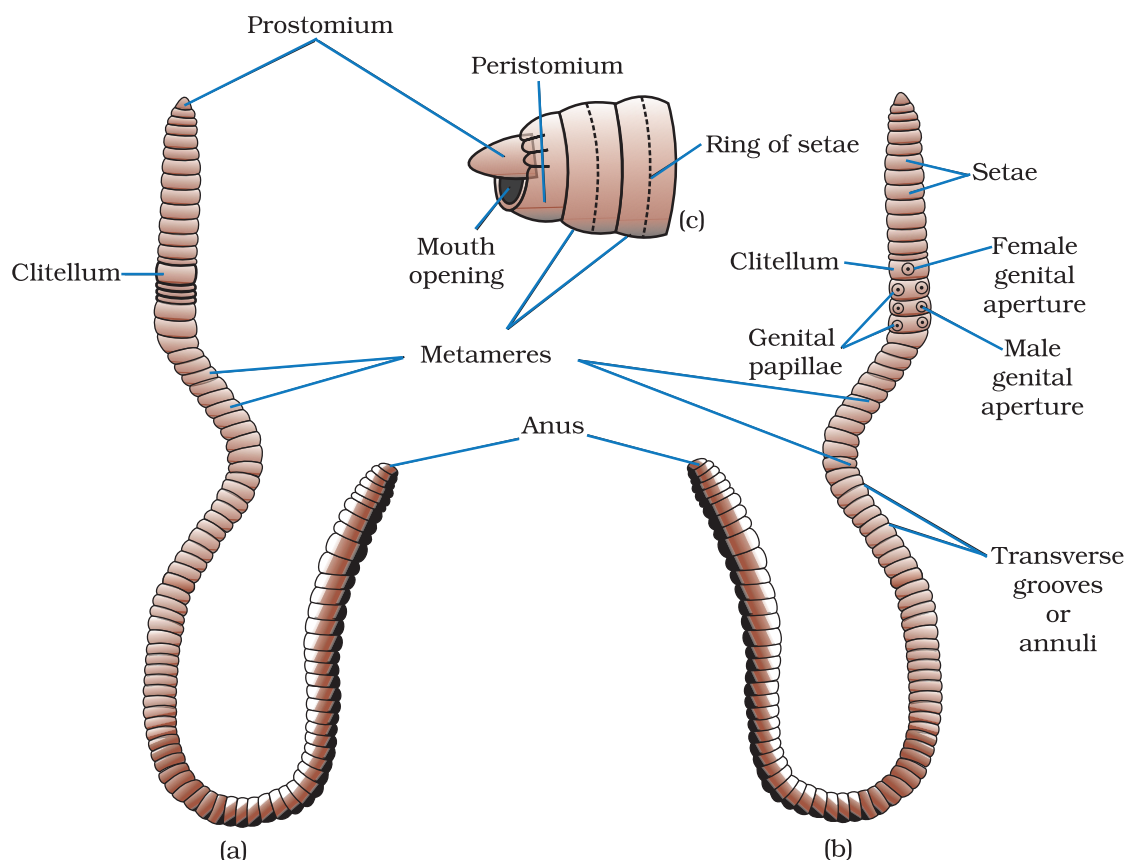


Fig. 3.13 Body of earthworm: (a) dorsal view (b) ventral view (c) lateral view showing mouth opening

lobe which serves as a covering for the mouth, the **prostomium**, which is not a true segment and over hangs the oral aperture. The prostomium bears some sense receptors. The first body segment is the **peristomium** (*buccal segment*) which surrounds the mouth and the prostomium. In a mature worm, the segments 14–16 are covered by a prominent dark band of glandular tissue called **clitellum** or **cingulum** (*L. clitellum*: a pack saddle; *cingulum*: a belt). The body is divisible into three prominent regions –the **pre-clitellar**, **clitellar** and **post-clitellar** regions (Figure 3.13). Four pairs of spermathecal apertures are situated on the ventro-lateral sides of the inter-segmental grooves, from 5th to 9th segments (5/6, 6/7, 7/8 and 8/9). A single female genital pore is present in the mid-ventral line of the 14th segment. A pair of male genital pores is present on the ventro-lateral sides of the 18th segment. Numerous minute pores called nephridiopores open on the surface of the body from the 3rd segment to the last. In each body segment, except the first, last (pygidium) and clitellum (in mature worms) there are rows of s-shaped chitinous **setae** or **chaetae** embedded in the epidermal pits (setal or setigerous sacs) in the middle of each segment. Setae can be protruded or retracted. They play an important role in locomotion.

II. Anatomy

The body wall of earthworm is covered externally by a thin non-cellular cuticle below which is the epidermis, two muscle layers (outer circular and inner longitudinal) and the innermost layer, the outer coelomic epithelium (parietal peritoneum). The epidermis is made up of a single layer of columnar epithelial cells which contain secretory gland cells. The coelomic epithelium is derived from the mesoderm and consists of the outer parietal (somatic) layer and inner splanchnic (visceral) layer. The space between the body wall and alimentary canal is called coelom (perivisceral cavity). It is a *schizocoelom* and is divided into compartments by septa filled with coelomic fluid. The coelomic fluid provides a '**hydrostatic skeleton**' for the earthworm during locomotion.

III. Digestive system

The alimentary canal is a straight tube and runs from the first to the last segment of the body (Figure 3.14). The mouth opens into the **buccal cavity** (1–3 segments) which leads into the muscular **pharynx** (4th segment). A small narrow tube, **oesophagus** (5–7 segments), continues into a muscular **gizzard** (8th segment). It helps in grinding the small particles of food the decaying leaves (**grinding mill**).

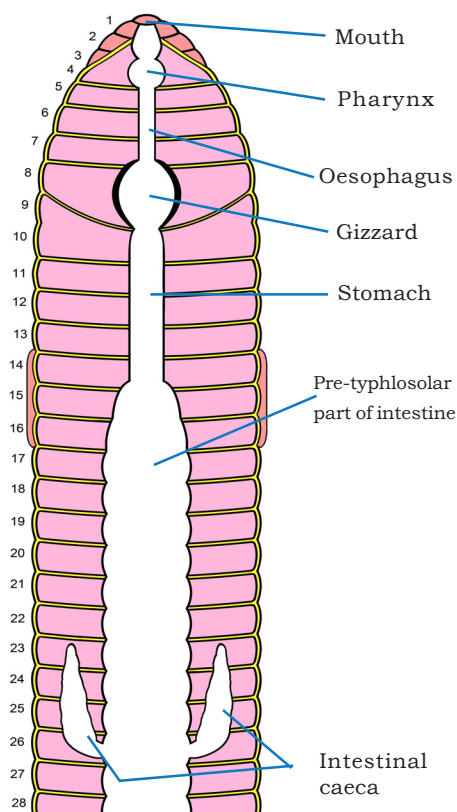


Fig. 3.14 Alimentary canal of earthworm

The stomach extends from the segments 9 to 14. The food of earthworm is decaying leaves and other organic matter mixed with the soil. Calciferous glands, present in the **stomach**, neutralise the humic acid present in the humus of the soil. The intestine starts from the 15th segment and continues till the last segment. A pair of short and conical intestinal caecae project from the intestine in the 26th segment. An internal median fold of the dorsal wall of the intestine called **typhlosole**, helping in increasing the area of absorption, is poorly developed in *Pheretima* (between the 26th and the **rectum**, which occupies the last 23 to 28 segments). The alimentary canal opens to the exterior by a small rounded aperture called anus. The ingested soil rich in organic matter passes through the digestive tract where digestive enzymes breakdown complex food into smaller absorbable units. These simpler molecules are absorbed through intestinal membranes and are utilised for various metabolic activities.

IV. Blood vascular system

Pheretima exhibits a **closed type** of blood vascular system, consisting of blood vessels, capillaries and 4 pairs of hearts. (Figure 3.15). Hearts connect the dorsal and ventral blood vessels. Due to closed circulatory system, blood is confined to the blood vessels. Contractions of the DBV keep blood circulating in the body. Smaller blood vessels supply blood to the gut, nerve cord and the body wall. **Blood glands** are present in the 4th, 5th and 6th segments. They produce blood cells and 'haemoglobin' which is 'dissolved in the plasma'. Blood cells are phagocytic in nature.

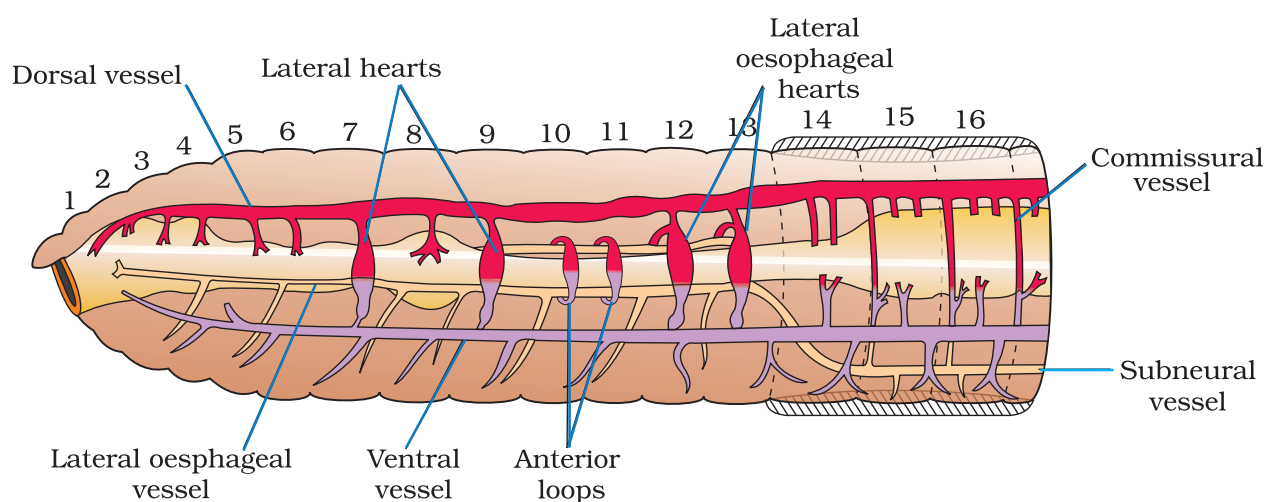


Fig. 3.15 Earthworm circulatory system

V. Respiratory system

Earthworms lack specialised respiratory structures. Respiratory exchange of gases occurs through the moist body surface by diffusion.

VI. Excretory system

The excretory organs occur as segmentally arranged coiled tubules called **nephridia** (sing: *nephridium*). They are of three types:

i) septal nephridia, present on both the sides of the inter-segmental septa of segments 15th/16th to the last. They open into the intestine

ii) integumentary nephridia, attached to the inner body wall from the 3rd segment to the last. They open to the exterior on the body surface by nephridiopores and

iii) pharyngeal nephridia, present as three paired tufts in the segments 4th, 5th and 6th (Figure 3.16). They open into the buccal cavity and pharynx. The different types of nephridia of *Pheretima* are fundamentally similar in structure. Nephridia with nephrostomes are called **open** nephridia (e.g. septal nephridia) and the nephridia which do not have nephrostomes are called **closed** nephridia (e.g. pharyngeal nephridia and integumentary nephridia). The nephridia that open to the exterior through nephridiopores are called '**exonephric nephridia**' and those opening into the gut are called '**enteronephric nephridia**'. Enteronephric nephridia have a role in **osmoregulation**/conservation/**homeostasis** of water. Earthworms are mostly *ureotelic* animals. The chief nitrogenous excretory waste is urea.

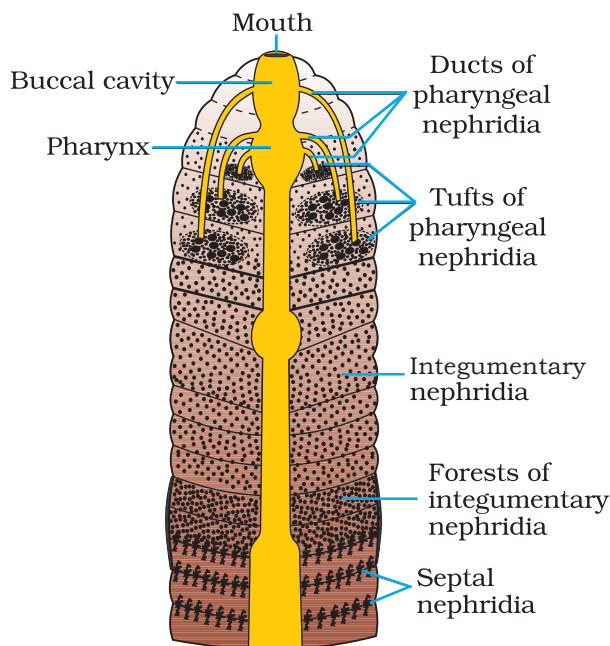


Fig. 3.16 Nephridial system in earthworm

VII. Nervous system

Nervous system consists of a **nerve ring** around the pharynx (in the 3rd and 4th segments) and a double ventral nerve cord. The cerebral ganglia (brain) along with other nerves in the ring integrate sensory input as well as command muscular responses of the body. Sensory system consists of receptor cells that receive light, touch and chemical stimuli. These sense organs are located mostly in the anterior part of the body.

VIII. Reproductive system

Earthworm is a hermaphrodite (bisexual) (Figure 3.17). There are two pairs of **testes** one pair each present in the 10th and 11th segments. Their **vasa deferentia** run up to

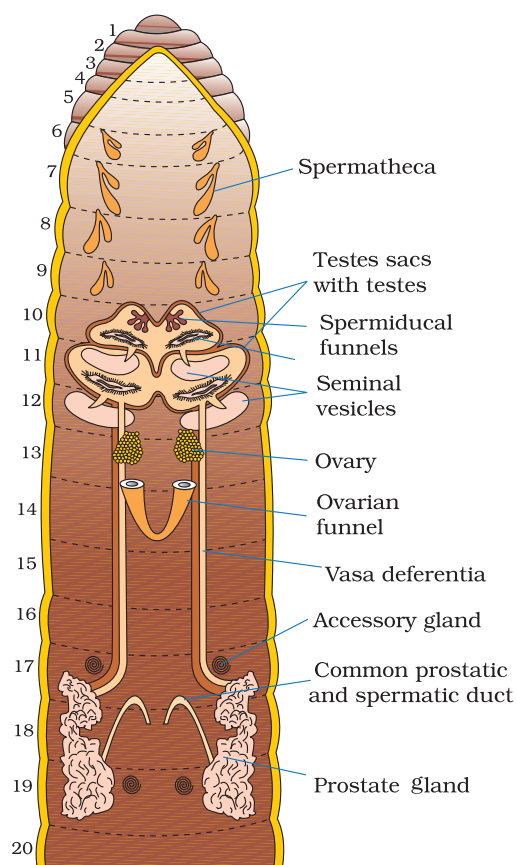


Fig. 3.17 Reproductive system of *Pheretima*

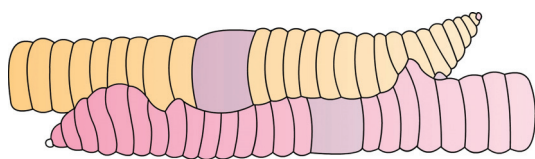


Fig. 3.18 Copulation

the 18th segment where they join the prostatic ducts. Two pairs of seminal vesicles present in the 11th and 12th segments are sacs in which spermatogonia mature into spermatozoa. The common prostatic and spermatic ducts open to the exterior by a pair of male genital pores on the ventro-lateral sides of the 18th segment. Two pairs of accessory glands are present one pair each in the 17th and 19th segments. Four pairs of **spermathecae** are located in the segments 6th to 9th (one pair in each segment). They receive and store spermatozoa (**spermatophore**) during copulation.

One pair of ovaries is attached to the posterior face of the inter-segmental septum of the 12th and 13th segments. *Oviducal funnels* are present beneath the ovaries and they continue into oviducts (14th segment). They join together and open to the exterior on the ventral side of the 14th segment by a single median female genital pore.

IX. Fertilization, cocoon formation and Copulation

Earthworm is protandrous. Mutual exchange of spermatozoa occurs between two mature worms during mating. They mate juxtaposing opposite gonadal openings, exchanging packets of sperms called **spermatophores** (Ref. NCERT book). Mature sperms and egg cells and nutritive fluid produced by the gland cells of the clitellum are deposited in **cocoons**. Fertilisation and development occur within the cocoons which are deposited in the moist soil. The cocoon contains the developing embryos. After about 3 weeks, each cocoon produces on average four earthworms (Ref: NCERT text book). Development is direct i.e., there is no larval stage.

X. Economic importance

Earthworms are known as **Friends of farmers** because they make burrows in the soil and make it porous, which helps in aeration of the soil. The process of enriching the fertility of soil by earthworms involves the process called **vermi-composting**. Earthworms are also used as bait in fishing.

3.7 PHYLUM-ARTHROPODA

- *Tagmatization of the body; jointed legs.*

3.7.1 General Characters

This is the **largest phylum** of the Kingdom: Animalia, and it includes the **largest Class** called **Insecta**. Over two-thirds of all named species on earth are arthropods accounting for 80% of the animal species (Figure 3.19 to 3.25). They are bilaterally symmetrical, triploblastic, metamerically segmented and coelomate (**schizocoelomate**) animals. The body of arthropods is covered by a chitinous exoskeleton as a protection and to prevent loss of water and it is periodically shed off by a process called **moulting** or **ecdysis** to allow growth of the body. The body is segmented and consists of *head*, *thorax* and *abdomen*. They have jointed appendages (arthros : joint; podium : foot). Muscles are striated and aid in rapid locomotion. (**Note: striated muscles appeared for the first time in evolution, in the arthropods**). Body cavity is a **haemocoel**. It is a cavity filled with haemolymph. It is formed by the fusion of embryonic blastocoel with some coelomic spaces. The coelom is reduced to the spaces around excretory organs and gonads. Respiratory organs are *gills*, *book gills*, *book lungs* or *tracheae*. Circulatory system is of **open type**. Heart is dorsal in position. In some (crustaceans and chelecerates) the haemolymph contains a ‘*copper*’ containing respiratory pigment called ‘**haemocyanin**’ dissolved in the haemolymph. Nervous system is of annelidan type consisting of a *nerve ring* (around oesophagus) and a double ventral nerve cord. Sensory organs like antennae, eyes (compound and simple), statocysts (organs of balance/ equilibrium) are present. Excretion takes place through **malpighian tubules**, **green glands**, **coxal glands**, etc. They are mostly dioecious (unisexual / gonochoric). Fertilisation is usually internal. They are mostly oviparous. Development may be direct or indirect. Life history includes one to many larval stages followed by metamorphosis. Examples: *Periplaneta*, *Palaemon*, *Cancer*, *Palamnaeus*, *Aranea*; **Economically important insects** – *Apis*, *Bombyx*, *Kerria* (*Laccifer*); **Vectors of common pathogens** – *Anopheles*, *Culex* and *Aedes*; **Gregarious pest** – *Locusta*; **Living fossil** – *Limulus*.



3.7.2 Classification of the phylum Arthropoda

Phylum **Arthropoda** is classified into **three sub-phyla**, **1. Trilobita** **2. Chelicerata** and **3. Mandibulata**.

Sub-phylum-1: Trilobita

They are extinct marine arthropods. Appendages are numerous, similar and *biramous*. Body is divided into a median and two lateral lobes by two prominent *longitudinal furrows*, hence the name trilobite.

Examples: *Triarthrus*, *Dalmanites*.



Fig. 3.19 Trilobite (fossil)

Sub-phylum-2: Chelicerata

Body consists of an anterior six segmented **cephalothorax** or **prosoma**, a posterior thirteen segmented **abdomen** or **opisthosoma** which is further divided into **mesosoma** and **metasoma**. Prosoma bears six pairs of appendages; the first pair is modified for feeding (**chelicerae**). Antennae and jaws (mandibles) are absent. Respiratory organs are **book-gills**, **book-lungs** and **tracheae**.

The sub-phylum **Chelicerata** is classified into **two classes**, namely **1. Xiphosura** and **2. Arachnida**.

Class-i: Xiphosura

They are marine organisms with branchial respiration; includes both **extinct** and **only three genera of extant** (living) animals. Prosoma is covered by broad horse-shoe shaped **carapace**; bears a pair of chelicerae and four pairs of walking legs and one pair

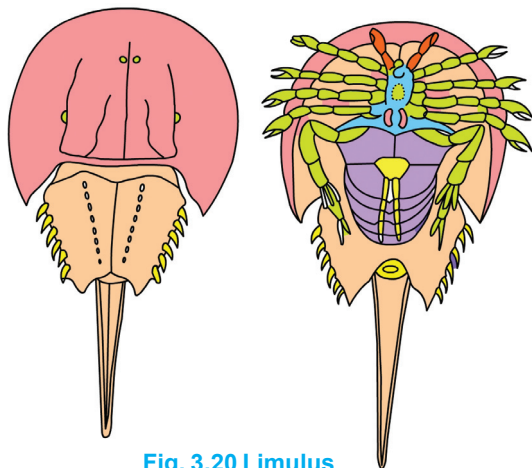


Fig. 3.20 Limulus

of '**pusher legs**'. Mesosoma bears six pairs of appendages, the first pair is fused to form **genital operculum** and the other five pairs are modified into **book-gills**. Metasoma ends in **telson**. Development includes **trilobite larva** in *Limulus*.

Examples: *Limulus* (king crab, a living fossil).

Class-ii: Arachnida

They are terrestrial; prosoma bears a pair of chelicerae, a pair of pedipalpi and four pairs of walking legs. Mesosomal appendages are modified into **book-lungs**. Four pairs of posterior abdominal appendages are modified into **spinnerets** in spiders. Respiratory organs are **book-lungs** (scorpions and some spiders), **tracheae** (some spiders) or **both** (some spiders). Respiratory pigment is '**copper**' containing **haemocyanin**. Excretory organs are **Malpighian tubules** and **coxal glands**. Development is **direct**; scorpions are **viviparous**.

Example: *Palamnaeus* (scorpion), *Aranea* (spider), *Sarcoptes* (itch mite).



Fig. 3.21 Scorpion

Sub-phylum-3: Mandibulata

They are aquatic (freshwater or marine) or terrestrial. Body is divided into two (cephalothorax and abdomen) or three (head, thorax and abdomen) parts. Head bears antennae, mandibles and maxillae. Respiratory organs are gills or tracheae.

The sub-phylum Mandibulata is classified into four classes **1. Crustacea** **2. Chilopoda** **3. Diplopoda** and **4. Insecta**.

Class-i: Crustacea

They are aquatic. Head and thorax fuse forming the **cephalothorax** (covered by chitinous **carapace**). In some the exoskeleton is hardened by calcium carbonate (crabs and lobsters). Cephalic region bears two pairs of antennae (**antennules** and **antennae** – **a unique feature**), one pair of mandibles and two pairs of maxillae. Thoracic and abdominal appendages are '**biramous**'. Respiratory organs are gills (branchiae). Excretory organs are **green glands** or **antennary glands**. Sense organs include **antennae**, **compound eyes**, **statocysts**, etc. Development is **indirect** and includes different larval forms.

Examples: *Palaemon* (freshwater prawn), *Cancer* (crab), *Balanus* (rock barnacle), *Sacculina* (root-headed barnacle), *Astacus* (cray fish), *Daphnia* (water flea).



Fig. 3.22 Prawn

Class-ii: Chilopoda

Chilopoda includes **centipedes** (*hundred leggers*). They are terrestrial, air-breathing, carnivorous animals. Body consists of head and trunk. Each segment of the trunk bears a pair of clawed appendages; first pair of trunk appendages bears **poison claws**. Respiratory structures are *tracheae*. Excretory organs are **Malpighian tubules**.

Examples: *Scolopendra*, *Scutigera*.



Fig. 3.23 Scolopendra

Class-iii: Diplopoda

Diplopoda includes **millipedes** (*thousand leggers*). They are terrestrial, air-breathing animals, feeding on decaying plant material. Body consists of head and trunk. Head bears paired antennae, mandibles and maxillae (maxillae are modified into **gnathochilarium**). Each trunk segment bears two pairs of legs. Respiratory organs are *tracheae*. Excretory organs are **Malpighian tubules**.

Examples: *Spirostreptus*, *Julus*.



Fig. 3.24 Spirostreptus

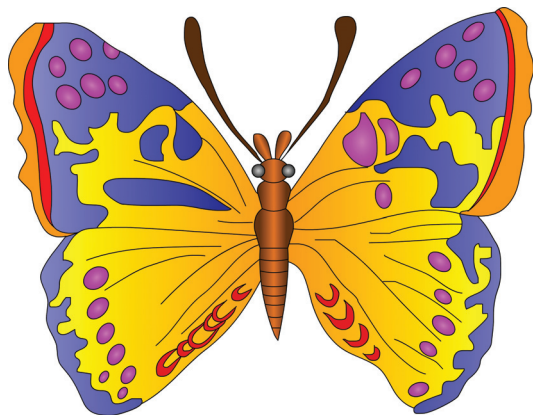


Fig. 3.25 Butterfly

Class-iv: Insecta

Insecta includes the most successful and the largest group of animals on the Earth. They are cosmopolitan, but absent from the marine habitat. Body consists of head, thorax and abdomen. Head bears paired compound eyes, antennae, mandibles, maxillae and an unpaired labium. Thorax consists of three segments and each segment bears a pair of *jointed walking legs*, hence this class is also referred to as **hexapoda**; thorax also bears one to two pairs of wings or wings may be absent (*Lepisma*). Respiratory organs are tracheae. Excretory organs are **Malpighian tubules**; they are **uricotelic** (excrete uric acid crystals as a **water conservation adaptation**). Development is mostly **indirect** with larval stages and **metamorphosis**.

Examples: *Periplaneta* (cockroach), *Musca* (house fly), *Lepisma* (silver fish).

3.8 PHYLUM-MOLLUSCA

- Unsegmented schizocoelomates

3.8.1 General Characters

Mollusca is the **second largest** phylum (Figures 3.26-3.28) in the Kingdom Animalia. Molluscs are terrestrial or aquatic (marine or fresh water). They are **bilaterally symmetrical** (some are asymmetrical, e.g. snail) and **coelomate** animals. Body is covered by a **calcareous shell** (mostly external, in some internal or reduced or absent) and is **un-segmented** with a distinct **head**, **muscular foot** and **visceral mass / visceral hump**. A soft and spongy layer of skin forms the **mantle** (*pallium*) over the visceral mass and secretes the shell. The space between the visceral mass and the mantle is called the '*mantle cavity*' in which comb like gills (*ctenidia*) are present. They have

Fig. 3.26 *Pila globosa*

respiratory and excretory functions (Ref: NCERT text book). The anterior head region has sensory tentacles. The body cavity is a **haemocoel**, and the true coelom is confined to the kidneys, gonads and pericardial space. The buccal cavity contains a file-like rasping organ called **radula** for feeding except in the bivalves. Except for cephalopods circulatory system is of *open type*; haemolymph contains a copper containing respiratory pigment, the **haemocyanin**. Nervous system consists of ganglia, commissures and connectives; sense organs are tentacles, eyes and **osphradium** (tests the purity of water – present in the bivalves and gastropods). Excretory organs are

metanephridia (commonly called *kidneys* or renal organs). They are usually dioecious (unisexual /gonochoric) and oviparous with indirect development which includes a characteristic ciliated **veliger larva** (*a modified trochophore*).

Examples: *Pila*, *Pinctada*, *Sepia*, *Loligo*, *Octopus*, *Aplysia*, *Dentalium*.

3.8.2 Classification of the phylum Mollusca

Phylum Mollusca is classified into seven classes, namely **1. Aplacophora** **2. Polyplacophora** **3. Monoplacophora** **4. Gastropoda** **5. Scaphopoda** **6. Pelecypoda** and **7. Cephalopoda**.

Class-1: Aplacophora (Gr. *a* : without; *placos* : plate; *pherein* : bearing)

They are primitive ‘worm like’ marine molluscs without mantle, shell, foot and nephridia. Head is poorly developed; a **radula** is present. Cuticle contains **calcareous spicules**. In some there is a *mid-ventral groove* which is homologous to the foot of the other molluscs.

Examples: *Neomenia*, *Chaetoderma*.

Class-2: Polyplacophora (Gr. *poly* : many; *placos* : plate; *pherein* : bearing)

This class includes **chitons**. They are bilaterally symmetrical and dorsoventrally flattened. Shell is dorsal and consists of **eight** transverse plates (*valves*). Foot is ventral, elongated and flat. Gills are 6 to 88 pairs. Development includes a trochophore larva.

Examples: *Chiton*, *Lepidopleurus*.

Class-3: Monoplacophora (Gr. *Mono* : single; *placos* : plate; *pherein* : bearing)

The members of this class were believed to have become extinct millions of years ago; *Neopilina* was collected from the deep sea off the Pacific coast of Costa Rica in 1952 by the oceanographic research vessel called **Galathea**. They are bilaterally symmetrical. Nephridia and gills show ‘**serial repetition**’ (some call it **internal segmentation**). Shell is single and plate like. Heart is unique with two pairs of atria opening into two ventricles.

Example: *Neopilina galathea*.

Class-4: Gastropoda (Gr. *Gaster* : belly; *podos* : foot)

Gastropoda is the largest and most diverse class of the phylum Mollusca; includes snails, slugs, limpets, etc.; mostly marine; some are freshwater or terrestrial. Shell is external, univalve and spirally coiled or internal (*Aplysia*) or absent (*Doris*). Head is distinct with eyes and tentacles. **Radula** occurs in the buccal cavity. Foot is ventral, flattened, muscular and helps in creeping. Respiratory organs are gills or lungs (pulmonary sacs) or both. During embryonic development, one side of the visceral mass grows faster than the other side; this uneven growth rotates the visceral mass up to 180°. This process is called ‘**torsion**’. They are mostly dioecious; development is indirect and includes **veliger larva**.

Examples: *Pila* (apple snail), *Aplysia* (sea hare), *Doris* (sea lemon).



Fig. 3.27 *Pila globosa* (shell)

Class-5: Scaphopoda (Gr. *Scapho* : boat, hollow shell; *podos* : foot)

They are marine and are commonly called '**tusk shells**' or '**tooth shells**' as the shell is tubular and '*tusk*' like, opening at both the ends. Foot is *cone* like and useful for digging. Eyes, tentacles, *ctenidia* (gills) and atria are absent. A large number of thread like structures called **captacula** help in the capture of food.

Examples: *Dentalium*, *Pulsellum*.

Class-6: Pelecypoda or Bivalvia or Lamellibranchiata (Gr. *pelekys* : axe, hatchet; *podos* : foot)

This class includes *mussels*, *oysters*, *clams*, etc. Body is laterally compressed and covered by a shell made up of two valves (Bivalvia). Head and sensory appendages are greatly reduced. Foot is '*wedge shaped*' (hence the name pelecypoda) and adapted for digging. Byssus threads occur in some (e.g. *Mytilus*) and help in attaching to the substratum. Radula is absent. They are '*suspension feeders*' or '*filter feeders*'. In some a '**crystalline style**' occurs in the stomach and helps in the digestion of starches. Respiratory organs are '*plate like*' gills (hence the name lamellibranchiata). They are dioecious (sexes are separate); development is indirect including larval forms such as **trochophore** and **veliger**; a **glochidium larva** occurs in freshwater mussels such as *Unio* (adapted for parasitic life on the gills of fish).

Examples: *Unio* (freshwater mussel), *Mytilus* (marine mussel), *Pinctada* (pearl oyster).

Class-7: Cephalopoda or Siphonopoda (Gr. *cephalo* : head; *podos* : foot)

This class includes *cuttle fishes*, *squids*, *octopuses*, *nautili* etc. Head is distinct with conspicuous eyes similar to those of vertebrates, a pair of horny beak like jaws and a radula occur in the buccal cavity. Shell may be external and multi-chambered (*Nautilus*) or internal (*Sepia*, *Loligo*) or absent (*Octopus*). The shell of *Sepia* is commonly called '**cuttle bone**' and that of *Loligo* is commonly called '**pen**'. Foot is modified into eight (*octopus*) or ten arms (*Sepia*, *Loligo*) provided with suckers present around the mouth and a part of the foot is modified into a '**siphon**' (useful in swift darting movements). Some possess **ink gland** and eject a cloud of ink to escape from the

predators (*defensive adaptation*). Ctenidia, atria and nephridia are *two* in **dibranchiates** (*Sepia*) and *four* in **tetrabranchiates** (*Nautilus*). Circulatory system is a **closed type** (*unique feature of cephalopoda*); heart has two to four atria and a ventricle. Nervous system is well developed with a well developed brain enclosed in a **cartilaginous cranium** (*brain case*). They are dioecious (sexes are separate); development is direct.

Examples: *Sepia* (cuttle fish), *Architeuthis* (giant squid - the largest living invertebrate), *Nautilus*, *Octopus* (devil fish).



Fig. 3.28 Octopus

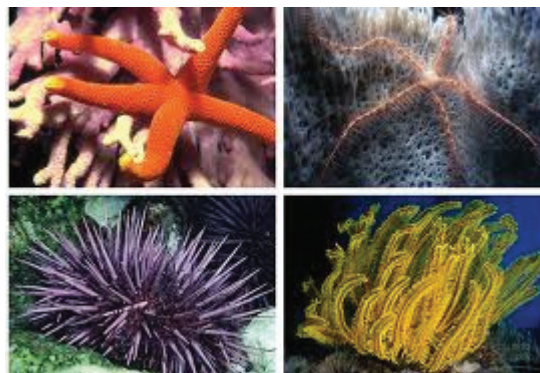
3.9 PHYLUM-ECHINODERMATA

- Deuterostomes with radial bodies

3.9.1 General Characters

The **echinoderms** (Gr. *echinos* : spiny; *dermos* : skin) are non-chordate deuterostomes and enterocoelomates. All are marine. The adult echinoderms are **radially symmetrical** (**pentamerous radial symmetry**), but the larvae are **bilaterally symmetrical**. Body wall is thick and body surface spiny; echinoderms possess a mesodermal endoskeleton of calcareous ossicles, which support the spines, hence the name 'Echinodermata' (*spiny bodied animals*). In some, the spines are modified into '**pedicellariae**'. The most distinctive feature of echinoderms is the presence of **water vascular system** or **ambulacral system** derived from the coelom which helps in locomotion, capture and transport of food and respiration. Water enters the ambulacral system through the *madreporite*. Circulatory system is of **open type**, without heart and blood vessels. Respiratory organs are '**papulae**' or '**dermal branchiae**' (thin outgrowths of the body wall between the spines), **cloacal respiratory trees** (*sea cucumbers*), etc. Special excretory organs are absent (excretion is by simple diffusion). Nervous system is poorly developed and brain is absent; sense organs are poorly developed. Sexes are separate (dioecious) but do not exhibit sexual dimorphism. Reproduction is sexual. Fertilisation is usually external. Development is indirect with a free-swimming **bilaterally symmetrical larva**. Power of regeneration is remarkable. Some exhibit **autotomy** (*auto* : self; *tome* : cut; self mutilation or self amputation) when captured by a predator, and the lost body parts are regenerated. A hypothetical larva, **dipleurula** is considered the ancestor of all echinoderms.

Examples: *Asterias*, *Echinus*, *Antedon*, *Cucumaria* and *Ophiura*.



3.9.2 Classification of the phylum Echinodermata

Phylum Echinodermata is classified into **two sub-phyla** namely **1. Pelmatozoa** and **2. Eleutherozoa**.

Sub-phylum-1: Pelmatozoa

They are sessile, attached to the substratum by long stalk (*sea lilies*) and some are free swimming (*feather stars*). Mouth and anus are present on the upper surface (oral surface). Madreporite is absent. Ambulacral grooves are open.

The sub-phylum **Pelmatozoa** includes **one class only 1. Crinoidea**

Class-1: Crinoidea (Gr. *crinon* : lily; *eidos* : form)

This class includes the oldest and most *primitive* forms. Some are stalk-less and free moving and others are stalked. Viscera is enclosed in a calcareous *test* called **theca**. Arms are **biramous** and bear **pinnules** and help in food capture. Mouth and anus are

present on the upper surface (oral surface). Ambulacral grooves are **open**. Tube feet do not bear suckers. Madreporite, spines and pedicellariae are absent. Some attach to the substratum by 'cirri'. Development includes **doliolaria larava**; it develops into a stalked, sessile **pentacrinoid larva** in feather stars.

Examples: *Ptilocrinus* (Sea lily with cirri), *Bathycrinus* (sea lily without cirri) and *Neometra* (feather star).

Sub-phylum-1: Eleutherozoa (Gr. *eleuthros* : free; *zoon* : animal)

They are free moving and do not have a stalk. Mouth (oral surface) is directed downwards. Madreporite is present. Ambulacral grooves are closed (except in *Asteroidea*).

The Sub-phylum **Eleutherozoa** is classified into **four classes 1. Asteroidea 2. Ophiuroidea 3. Echinoidea** and **4. Holothuroidea**.

Class-1: Asteroidea (Gr. *aster* : star; *eidos* : form)



Fig. 3.29 Starfish

This class includes sea stars (*star fishes*). The arms are not clearly demarcated from the central 'disc' of the body. **Ambulacral grooves** are open; tube feet bear suckers. Pedicellariae are two jawed and are useful in food capture, protection and cleaning the surface of the body. Madreporite is aboral (dorsal side) in position. Respiratory organs are **papulae** or **dermal branchiae**. Development includes **bipinnaria** and **brachiolaria larvae**.

Examples: *Asterias*, *Astropecten*, *Pentaceros*.

Class-2: Ophiuroidea (Gr. *ophiuros* : serpent tail; *eidos* : form)

This class includes *brittle stars*, *serpent stars*, *basket stars*, etc. Arms are long and clearly demarcated from the central disc of the body; arms are flexible, fragile and '**branched**' in some (*Gorgonocephalus*). They move by serpentine lashing of the arms.

Ambulacral grooves are closed; tube feet lack suckers. Madreporite is on the oral surface. Pedicellariae are absent. Anus is absent. Development includes **ophiopluteus larava**.

Examples: *Ophiothrix* (spiny brittle star), *Gorgonocephalus* (basket star).



Fig. 3.30 Brittle star

Class-3: Echinoidea (Gr. *echinos* : hedgehog; *eidos* : form)

This class includes 'sea urchins' such as, *heart urchins*, *sand dollars*, *sea biscuits*, *cake urchins* etc. The body is ovoid (*sea urchin*) or discoid (*sand dollars*) and covered with movable spines. Arms are absent;

tube feet bear suckers. Calcareous ossicles of the body unite to form a rigid **test** or **corona** or **case**. Madreporite and anus are aboral in position. Ambulacral grooves are closed. Pedicellariae are three jawed. In the mouth of sea urchin a complex five jawed masticatory apparatus called **Aristotle's lantern** is present (absent in the heart urchins). Development includes **echinopluteus larva**.

Examples: *Echinus* (sea urchin), *Echinocardium* (heart urchin), *Echinodiscus* (sand dollar), *Clypeaster* (cake urchin).

Class-4: Holothuroidea (Gr. *holothurian* : water polyp or sea cucumber; *eidos*:form)

This class includes sea cucumbers. Body is elongated in the oro-aboral axis; body wall is leathery (coriaceous) and dermis contains loose spicules. Arms, spines and pedicellariae are absent. Mouth is surrounded by retractile tentacles (modified tube feet useful for feeding). Ambulacral grooves are 'closed'; tube feet bear suckers. Madreporite is *internal* (occurs in coelom). Respiratory organs are a pair of cloacal '**respiratory trees**'. Development is indirect and includes **auricularia larva**.

Examples: *Holothuria*, *Synapta*, *Thyone*.

3.10 PHYLUM - HEMICHORDATA

- Half chordates

3.10.1 General Characters

Hemichordata was earlier considered a sub-phylum under phylum Chordata. But now it is placed as a separate phylum under the non-chordates. This phylum consists of a small group of **worm-like** marine animals. They are **bilaterally symmetrical**, **triploblastic** and **coelomate (enterocoelous) animals**. The body is cylindrical and is composed of an anterior **proboscis**, a **collar** and a long **trunk** (Figure 3.31). A median **buccal diverticulum**, the **stomochord** (once believed to be the notochord) extends into the proboscis. Circulatory system is of open type with a dorsal heart. Respiration takes place through paired gill slits opening into the pharynx. Proboscis gland acts as the excretory organ. Development is indirect and includes a **tornaria larva**.

Examples: *Balanoglossus* (acorn worm), *Saccoglossus*.

3.10.2 Classification of the phylum Hemichordata

Phylum : Hemichordata is classified into **two classes**,

1. Enteropneusta and **2. Pterobranchia**.

Class-1: Enteropneusta (Gr. *enteron* : gut; *pneustos* : breathed)

They are commonly called *acorn worms*. They are free, solitary and burrowing animals. Development includes a free swimming **tornaria larva**.

Examples: *Balanoglossus* (acorn worm), *Ptychodera*.

Class-2: Pterobranchia (Gr. *pteron*: feather or wing; *branchion* : gill)

They are sedentary, colonial (*Rhabdopleura*), and tube dwelling animals.

Examples: *Rhabdopleura*, *Cephalodiscus*.

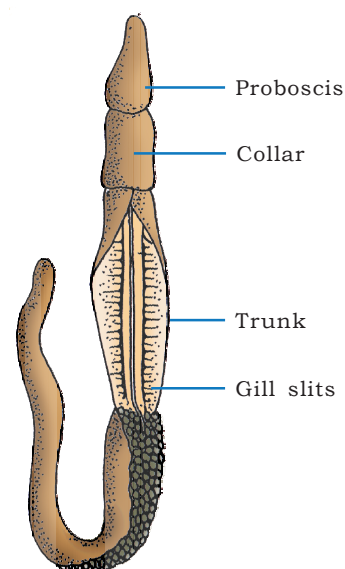


Fig. 3.31 Balanoglossus

GLOSSARY

Amphids: They are the cuticular depressions present on the lips surrounding the mouth in the nematodes. They are well developed in free living nematodes and serve as chemoreceptors.

Autotomy: It is the process of voluntary breaking of the injured part of the body (self mutilation or amputation). It is a device developed to protect the body from the enemies and parasites as seen in the echinoderms .

Bothridia: The leaf like organs present on the scolex of certain cestodes. They act as the organs of attachment to the body parts of the host.

Cercaria: It is a free swimming larva consisting of an ovoid body and a tail. It occurs in the life history of liver flukes.

Choanocyte: It is a specialized flagellated cell with a cup shaped collar around the base of the flagellum. It maintains the current of water through the body.

Clitellum: It is a belt or *girdle* shaped thickening of the skin, in a specific region of the body (14 to 16 segments in *Pheretima*). It secretes the cocoon and albumen (food for the developing young one), during the breeding season.

Cocoon: It is a bag like structure secreted by the clitellum. Eggs and sperms are deposited into it. Fertilization and development occurs within the cocoon.

Comb plates: They are ciliated plates helping in locomotion in the ctenophores.

Ctenidia: They are the respiratory organs of molluscs. Each ctenidium (gill) consists of a central axis and one or two rows of filaments.

Flame cell: It is a hollow cell containing a tuft of cilia in the inner space. It is excretory and osmoregulatory in function. Flame cell is a primitive type of excretory organ (protonephridium).

Gemmules: They are the '*internal buds*' of some sponges aiding in asexual reproduction. They also help tide over unfavourable conditions.

Juvenile: It is an immature young form resembling the adult in all aspects.

Madreporite: It is a circular porous plate present in the body of most echinoderms. Through its pores sea water enters the water vascular system.

Mesenteries: They are vertical partitions dividing the coelenteron of sea anemones. They are developed as inward folds of the endoderm. They bear cnidoblasts.

Miracidium: It is a free swimming larva of liver fluke. It contains ciliated epidermis, penetration glands, etc. It transforms into the sporocyst stage in the body of freshwater snail , which is the invertebrate host of liver fluke.

Moulting or Ecdysis: It is the process of casting off the outer body wall. During development of nematode larvae, the cuticle is cast off 4 times.

Muller's larva: It is the characteristic larva of turbellarians. It bears an apical tuft of cilia.

Nephridia: They are highly coiled excretory organs derived from the ectoderm.

Osculum: It is a large opening present at the free end of a sponge body. Water from the spongocoel flows out through the osculum.

Ostia: They are numerous openings present on the body of a sponge. Water is drawn in through the ostia and sent out through the osculum (choanocytes play an important role in drawing in water).

Parapodia: They are hollow biramous appendages of polychaetes. They bear tufts of many setae and help in locomotion.

Parenchymula: It is the larva of sponges such as *Clathrina*. It is similar to the planula of cnidarians. It is covered with cilia all over.

Pedicellariae: They are 'forceps like structures' (modified spines) in the body of sea stars and sea urchins. They remove foreign particles that may settle on the body surface and thus help in keeping the body surface clean.

Phasmids: They occur in some nematodes. They are well developed in parasitic nematodes and are glandulo-sensory in function.

Planula: It is a ciliated, double walled, free swimming larva characteristic of a cnidarian.

Polyembryony: Formation of several young ones from a single zygote. It occurs during the formation of redia and cercaria larvae in the life history of a liver fluke.

Polyp or Hydranth: It is a flower shaped or hydra shaped zooid of the colonial cnidarians such as *Obelia*. It bears several tentacles around the mouth. It is a nutritive zooid.

Radula: It is a thin ribbon shaped body lying in the buccal cavity of molluscs. It consists of a median stalk and two lateral lobes. Each lobe bears many rows of chitinous teeth. Radula works like a 'file' and rasps the vegetable food.

Redia: It is an elongated larva of liver fluke produced by the sporocyst asexually/ parthenogenetically. It gives rise to the second generation of rediae or the cercaria larvae.

Rhabdites: They are rod shaped bodies present in the epidermal cells of turbellarians. On discharge they form a slimy (mucus) substance helping in locomotion by smoothening the path of gliding movement and also in defence.

Setae: They are small, *f*-shaped, chitinous structures present in the pits (*setal sacs*) of the body wall of earthworms. They aid in locomotion. In some setae are modified into '*penial setae*' and help in copulation.

Spicules of a sponge: They are the supporting structures of the body secreted by *scleroblasts*. They are of different shapes and sizes. They are made up of calcium carbonate or silica.

Spongocoel: It is the large central cavity of a sponge body.

Statocyst: Statocyst is a sense organ concerned with the maintenance of equilibrium of the body.

Syncytium: It is a multinucleate mass of protoplasm without any division into cells. The tegument of flatworms is syncytial.

Trichemella: It is a larval form characteristic of the hexactinellid sponges.

Trochophore: It is the characteristic larva of annelids. It is 'top' shaped and bears one or more bands of cilia, an apical tuft, an eye etc. It contains coelom and nephridia. It is believed to be the forerunner of higher invertebrates.

Zooid: It is an individual member of a 'coelenterate' colony. Different types of zooids serve different functions (division of labour).

VERY SHORT ANSWER TYPE QUESTIONS

1. Name the cells and system peculiar to porifera.
2. Which structures form the skeleton of a sponge? What are the chemicals involved in the formation of these structures?
3. What are the functions of canal system of sponges?
4. What are the two chief morphological 'bodyforms' of cnidarians? What are their chief functions?
5. What is metagenesis? Animals belonging to which phylum exhibit metagenesis?
6. What is the cnidarian group with quantitatively/relatively large mesoglea? What is the significance of such a well developed mesoglea pertaining to the aquatic life of that group?
7. What is the chief difference between the hydrozoans and the rest of the cnidarians regarding the germinal layer(s) in which its 'defensive structures or cells of defence' occur?
8. What are the excretory cells of flatworms called? What is the other important function of these specialized cells?
9. Distinguish between amphids and phasmids.
10. What is the essential difference between a 'flat worm' and a 'round worm' with reference to the perivisceral area of their bodies?
11. How do you account for the origin of the perivisceral space in the body of a nematode and an annelid?
12. What is metamerism? What is the essential difference between the mode of formation of individual morphological body units of a tapeworm and those of an earthworm?
13. How do you distinguish a 'hirudinean' from the rest of the annelids, based on the morphological features pertaining to metamerism? How does the coelom of a leech differ from the coelom of an earthworm with reference to its contents?
14. What do you call the locomotor structures of *Nereis*? Why is *Nereis* called a polychaete?
15. What is botryoidal tissue?
16. What is the difference between the epidermis of a nematode and that of an annelid? How does a nematode differ from an annelid with reference to the musculature of the body wall?
17. What do you call the first and second pairs of cephalic appendages of a scorpion?
18. What is the uniqueness about the first two pairs of cephalic appendages of a crustacean compared to those of the other extant arthropods?
19. What is the sub-phylum to which 'ticks' and 'mites' belong? How do you distinguish them from the insects with reference to their walking legs?
20. What are the respiratory structures of *Limulus* and *Palamnaeus* respectively?
21. What are antennae? What is the arthropod group without antennae?
22. What do you call the perivisceral cavity of an arthropod? Where from is it derived during development?
23. Which arthropod, you have studied, is called a 'living fossil'? Name its respiratory organs?
24. How do you identify a *Chiton* from its external appearance? How many pairs of gills help in the respiration of *Chiton*?

25. What is the function of radula? Give the name of the group of molluscs which do not possess a radula?
26. What is the other name for the gill of a mollusc? What is the function of osphradium?
27. What is Aristotle's lantern? Give one example of an animal possessing it?
28. What is the essential difference between the larvae and adults of echinoderms, symmetry wise?
29. What are blood glands in *Pheretima*?
30. What are spermathecae on the body of *pheretima*?

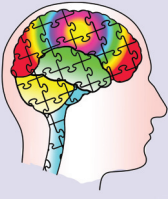
SHORT ANSWER TYPE QUESTIONS

1. Write short notes on the salient features of the anthozoans.
2. What is the class to which the flukes belong? Write short notes on the chief characters of that group?
3. What are the salient features exhibited by polychaetes?
4. How do the hirudineans differ from the polychaetes and oligochaetes?
5. What are the chief characters of the crustaceans?
6. Mention the general characters of Arachnida.
7. Compare briefly a centipede and a millipede.
8. Cephalopods show several unique or advanced features when compared to the other molluscs. Discuss briefly.
9. What are the salient features of pelecypoda.
10. What are the salient features of the echinoids?

11. Mention the salient features of Holothuroidea.
12. What is the function of nephridia?
13. Give an account of the hearts in the circulatory system of *Pheretima*.
14. Name the phylum to which comb jellies belong to and mention its important features.
15. Give a brief account of hemichordates.

LONG ANSWER TYPE QUESTIONS

1. Draw a labelled diagram of the reproductive organs of *Pheretima*.
2. Describe the digestive system and process of digestion in *Pheretima*.



For ignited minds

- Not For Evaluation

1. Why do we call *Limulus*, a 'Living fossil'?
2. How do you distinguish a '**King crab**' from a '**scorpion**' based on a morphologically distinctive character, on the ventral aspect of the prosomas of the two animals.
3. When there is no respiratory pigment in the blood of an insect or a centipede or a millipede, why should there be a pigment in the blood of a scorpion?
4. Blood flows in our body. Similarly blood flows in the body of cockroach also (in spite of the open circulatory system, it has). How come that the blood of cockroach is mostly referred to as 'haemolymph'?
5. There is a perivisceral cavity in the body of a nematode almost similar to that seen in the body of an earthworm. In such a case, why do you call it a 'pseudocoel' instead of 'coelom'.
6. "All non chordates are 'schizocoelomates' and chordates are 'enterocoelomates'. Do you agree with the above statement? Give reasons for your positive or negative answer.
7. "All schizocoelomates are protostomeates, while all protostomeates are not schizocoelomates". Do you think the statement is correct? If so give reasons. If not, how do you defend.
8. Echinoderms show 'pentamerous radial symmetry'. Yet, they are classified under the Grade Bilateria. Does it not look anomalous to you? What is your suggestion to taxonomists? Do you want them to reclassify the animal world?
9. In almost all animals of the animal kingdom, gonads arise from the germinal layer called mesoderm. However, some animals /animal groups differ. Can you name them and explain the origin of germ cells in them.
10. There is a maximum number of four chambers in the heart of a vertebrate. There are invertebrates with more number of atria and ventricles. Can you give examples for such organisms.